Ethan W. Todd

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Education

2023 – Present (Fall) — **Candidate, Ph.D. Physics**, University of Michigan, Ann Arbor, MI
Advisors: Prof. Igor Jovanovic, Prof. James Wells
2025 (Winter) M.S. Physics

2020 – 2023 (Fall – Spring) — **B.S. Physics**, *Summa Cum Laude*, Florida State University, Tallahassee, FL
Honors Thesis: Study of Displaced Vertex Tagging with the CMS Experiment
Advisor: Prof. Ted Kolberg

2018 – 2020 (Fall – Spring) — **Foundational Coursework**, Emory University, Atlanta, GA

Publications

4. **E. Todd**, V. Fondement, and I. Jovanovic, "Particle Identification in YAP:Ce," *under review*, 2025

3. C. Graham, **E. Todd**, C. Wilson, et al., "Light Output-Dependent Integration Gating for Optimized Pulse Shape Discrimination," *under review*, 2025

2. **CMS HGCAL Collaboration**, "Leakage current of high-fluence neutron-irradiated 8" silicon sensors for the CMS Endcap Calorimeter Upgrade," *under review*, 2025. [Online]. Available: `https://arxiv.org/abs/2510.17280`

1. K. Yocum, H. Smith, **E. Todd**, et al., "Millimeter/Submillimeter Spectroscopic Detection of Desorbed Ices: A New Technique in Laboratory Astrochemistry," *The Journal of Physical Chemistry A*, vol. 123, no. 40, pp. 8702–8708, 2019, ISSN: 10.1021/acs.jpca.9b04587

Research Experience

2024 – Present (Summer) — **Graduate Research Assistant**, University of Michigan, Ann Arbor, MI
Advisor: Prof. Igor Jovanovic
- Developed machine learning (ML) based particle identification techniques for inorganic scintillators in which traditional techniques, *i.e.* charge comparison, are ineffective.
- Simulated detection efficiency of gamma rays emitted following thermal neutron capture in lithium and germanium samples in preparation for experiments probing low energy nuclear recoil physics via gamma-ray tagging.

2022 – 2023 (Summer – Summer) — **Laboratory Assistant**, Florida State University, Tallahassee, FL
Advisor: Prof. Rachel Yohay
- Conducted process quality control (PQC) testing for the CMS High Granularity Calorimeter (HGCAL) to monitor the quality/stability of the silicon sensor production process.
- Analyzed and presented PQC results to the HGCAL sensor testing group.
- Helped create and implement a standardized testing procedure at FSU.

2021 – 2023 (Fall – Spring) — **Undergraduate Research Assistant**, Florida State University, Tallahassee, FL
Advisor: Prof. Ted Kolberg
- Participated in a deep neural network ML based search for beyond the Standard Model long-lived particles using the CMS Experiment.
- Trained ML models on Monte Carlo data with varying signal (mass, lifetime) and background (t, b physics) to verify their ability to identify displaced vertices independently of the particular physics.
- Investigated the most effective parameters of ML models; verified physics-based expectations.

2019 – 2019 (Summer – Fall) — **Summer Undergraduate Research Experience Fellow**, Emory University, Atlanta, GA
Advisor: Prof. Susanna Widicus Weaver/Dr. Katarina Yocum
- Conducted proof-of-concept millimeter/submillimeter spectroscopic measurements of H_2O and D_2O binding energy, sublimation enthalpy, and sublimation entropy.
- Assisted in data collection, analysis, and literature review.

2018 Fall – 2020 Fall **Undergraduate Research Assistant**, Emory University, Atlanta, GA
Advisor: Prof. Susanna Widicus Weaver
- Performed millimeter/submillimeter spectroscopic measurements of desorbed astrophysical ice analogs.
- Analyzed broadband molecular line surveys using the GOBASIC software package.

Teaching Experience

2025 Fall - 2026 Winter **Lead Graduate Student Instructor**, University of Michigan, Ann Arbor, MI
- Oversee 30 graduate student instructors teaching general physics laboratories with a total enrollment of 1800 undergraduate students.
- Edit and maintain the laboratory manuals and Jupyter notebook worksheets used in the course.
- Manage the course information and organization via Canvas.
- Arbitrate student and graduate student instructor concerns.

2023 Fall – 2025 Spring **Graduate Student Instructor**, University of Michigan, Ann Arbor, MI
- 2025 Spring Physics 251, Fundamental Physics for the Life Sciences II Lab
- 2024 Fall Physics 121, Physics for Architects Lab
- 2023 Fall – 2024 Winter Physics 241, General Physics II Lab

2022 Spring **Physics Learning Assistant**, Florida State University, Tallahassee, FL
PHY 2049C, General Physics B

2019 Fall **Laboratory Teaching Assistant**, Emory University, Atlanta, GA
CHEM 150L, Structure and Properties Lab

Honors, Awards, and Grants

2025 Fall Rackham Graduate Student Research Grant *Awarded to conduct research at the Paul Scherrer Institute on the ionization yield of high-purity germanium detectors.*

2023 Spring Phi Beta Kappa Society

2023 Spring Joseph Lanutti Undergraduate Research Award *Awarded for third place in the FSU physics departmental poster presentation contest.*

2022 Fall Evelyn and John Baugh Research Presentation Scholarship *Awarded to travel and present research at SESAPS.*

2022 Spring Anna Runyan Undergraduate Endowment *Awarded for outstanding academic success in physics.*

Computing

Languages

Python:
- NumPy
- SciPy
- PyROOT/uproot
- Scikit-Learn

Nix:
- Custom nixpkgs
- NixOS configuration

C++:
- ROOT
- GEANT4

LaTeX:
- Self-explanatory...
- Beamer

Operating Systems Linux (NixOS, Debian/Ubuntu, Arch Linux), MacOS

Skills

Programming	C++: ROOT, GEANT4 Python: NumPy, SciPy, Scikit-Learn, PyROOT/uproot C: (modify and build Chromium EC firmware and coreboot/edk2 firmware).
Software	**GEANT4** (Monte Carlo radiation transport), **ROOT** (data analysis), **Nix/NixOS** (package management and system configuration), $\LaTeX$ (including Beamer).
Hardware	Use of CAEN digitizers and associated software (CoMPASS, wavedump) for data acquisition; soldering and electronics assembly for experimental setups.
Operating Systems	Linux (NixOS, Debian/Ubuntu, Arch Linux), macOS.

Strong preference for open-source software and hardware.

Presentations

4. **E. Todd**, V. Fondement, and I. Jovanovic, "Particle Identification in Fast Inorganic Scintillators via Machine Learning," *Fall Meeting of the Eastern Great Lakes Section of the American Physical Society, Ypsilanti, MI*, 2025

3. **E. Todd** and I. Jovanovic, "Investigation of Anomalously High Ionization Produced by Low Energy Nuclear Recoils in Germanium (poster)," *DOE NNSA Office of Defense Nuclear Nonproliferation R&D University Program Review, Gainesville, FL*, 2025

2. **E. Todd**, A. A. Kadhim, N. Bower, et al., "Process Quality Control for HGCAL at Florida State University (poster)," *89th Annual Meeting of the Southeastern Section of the American Physical Society, Oxford, MS*, 2022

1. **E. Todd**, K. Yocum, P. Gerakines, et al., "A Novel Use of Rotational Spectroscopy for Studying Characteristics of Desorbed Interstellar Ice Analogs (poster)," *Emory SURE 2019 Symposium, Atlanta, GA*, 2019